

QUICK ACTIVITY 2: CANTEEN GROUP ORDER

PROBLEM ANALYSIS: A canteen needs to calculate the final bill for a group order of rice meals and drinks. The cost depends on the number of the rice meals, drinks, the discount of the rice meals, the packaging fee, and the payment.

DECOMPOSITION

- ① Read the meals, drinks, and payment
- ② Compute the meal subtotal only
- ③ Compute the 10% discount on the meal subtotal only.
- ④ Subtract the discount from the meal subtotal.
- ⑤ Compute the drink subtotal
- ⑥ Compute the packaging fee
- ⑦ Add the discounted meals, drinks, and packaging fee for the final bill.
- ⑧ Compute the change.
- ⑨ Display the final bill and change.

PSEUDOCODE

START

```

READ meals, drinks, payments
mealSubtotal ← meals × 85
discount ← mealSubtotal × 0.10
discountedMeals ← mealSubtotal - discount
drinkSubtotal ← drinks × 25
packagingFee ← meals × 5
finalBill ← discountedMeals + drinkSubtotal + packagingFee
change ← payment - finalBill
DISPLAY finalBill, change
    
```

END

DRY RUNS

TEST CASE 1:

3 rice meals, 4 drinks, ₱500

STEP	VARIABLE	CALCULATION	RESULT
1	mealSubtotal	$3 \times ₱85$	₱255
2	discount	$₱255 \times 0.10$	₱25.50
3	discountedMeals	$₱255 - ₱25.50$	₱229.50
4	drinkSubtotal	$4 \times ₱25$	₱100.00
5	packagingFee	$3 \times ₱5$	₱15
6	finalBill	$₱229.50 + ₱100 + ₱15$	₱344.50
7	change	$₱500 - ₱344.50$	₱155.50

TEST CASE 2:

5 rice meals, 3 drinks, ₱600

STEP	VARIABLE	CALCULATION	RESULTS
1	mealSubtotal	$5 \times ₱85$	₱425.00
2	discount	$₱425 \times 0.10$	₱42.50
3	discountedMeals	$₱425 - ₱42.50$	₱382.50
4	drinkSubtotal	$3 \times ₱25$	₱75.00
5	packagingFee	$5 \times ₱5$	₱25
6	finalBill	$₱382.50 + ₱75 + ₱25$	₱482.50
7	change	$₱600 - ₱482.50$	₱117.50